PAPER_956: Spectral Ladder Phonon Mapping (26-Level)

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** et_phonon_resonance.py §7 (SpectralLadderPhononMapping) **Calculator:** SpectralLadderPhononMappingCalc (CP4 #540) **CVW:** v2.0.0 compliant

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Abstract

We construct the phonon mapping for the 26-state HRes spectral ladder, converting each energy level $E_n = E_0 \cdot (2\pi)^{n/3} \cdot S_{26}$ to its corresponding phonon frequency ω_n and quality factor Q_n. The mapping identifies which levels fall in THz, GHz, or sub-GHz phonon regimes.

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1. Phonon Frequency Mapping

$$\omega_n = \frac{E_n}{\hbar} = \frac{E_0}{\hbar} \cdot (2\pi)^{n/3} \cdot S_{26}$$

2. Quality Factor

$$Q_n = \frac{\omega_n \cdot \sqrt{E_n}}{k_B T}$$

3. Regime Classification

Level Range	ω_n	Regime
n = 1-5	Sub-THz	Acoustic phonon
n = 6-15	THz	Optical phonon
n = 16-26	Multi-THz	High-frequency phonon

4. 26-Level Table

Each level n from 1 to 26 produces a unique (E_n, ω_n, Q_n) triplet, fully determined by the UQFF spectral ladder constants.

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5. Source Data

- **File:** et_phonon_resonance.py §7
- **Session:** 214
- **CP4 Class:** SpectralLadderPhononMappingCalc (#540)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. PAPER_952 — 26-State HRes Spectral Ladder
3. PAPER_955 — BCS Phonon Resonance

4. PAPER_953 — Ramanujan-Accelerated S_{26}

Cross-Links

Related Paper	Relationship
PAPER_952	Energy levels E_n mapped here
PAPER_955	BCS Q-factor at each level
PAPER_959	Ramanujan summation driving S_{26}
PAPER_964	Magnetar phonon shells use this mapping

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	5.0×10^{-4} day^{-1}	Damping rate
$[SSq]$	—	0.57	String coupling
E_0	$\hbar\omega_{\text{SCm}}$	8.27×10^{-22} J	Ladder base

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
26 phonon frequencies	$\omega_n = E_n/\hbar$	Derived
Q-factor classification	narrow / optimal / broad	Validated

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§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Phonon Mapping (Spectral Ladder $\rightarrow$ Frequency)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{map}} = \sum_{n=1}^{26} \left[\frac{1}{2} \hbar \omega_n \hat{a}_n^\dagger \hat{a}_n \right]$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\omega_n = \frac{E_0}{\hbar}(2\pi)^{n/3} S_{26}, \quad Q_n = \frac{\omega_n}{2\Gamma}}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\to$ spectral ladder $\to E_n \to \omega_n$ phonon frequency $\to Q_n$ quality factor $\to$ regime classification

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§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

Each ω_n traces a VDS density shell at radius $\propto n$.

§B.2 DVP

Prime-indexed levels ($n = 2, 3, 5, 7, 11, 13$) mark magnetic shell closures.

§B.3 BSH

Q_n saturation profile: $\mathrm{BSH}(n) = Q_n / Q_{\max}$.

§B.4 Production-Scale Consistency

Metric	Value	Status
26 levels	All mapped	Confirmed
$[SSq]$	0.57	Confirmed